

Machine Learning Engineer – Assessment

Spotter.AI

Muhammad Ayan BUTT

Ayanbutt1013@gmail.com

1. Dataset Preprocessing and Preparation (data_prepare.py)

The dataset (train-test.csv) was first processed and prepared before any model development:

- Loaded the dataset containing 48,000 freight records
- Non-predictive features/ columns were removed from the dataset
 - load_id
 - pickup
 - delivery
- Date feature was converted into following numerical features:
 - date_month
 - date_weekday
- Two additional features were created as a result of feature engineering:
 - $\text{delta_lat} = \text{delivery_lat} - \text{pickup_lat}$
 - $\text{delta_lon} = \text{delivery_lon} - \text{pickup_lon}$
- Missing n/a values were filled using mean (in case of numeric) and mode (in case of categorical)
- One-hot encoding was applied to “equipment”
- Standardized numerical features using StandardScaler.
- Applied a “log1p transformation” to the target variable (posted_rate) before training.

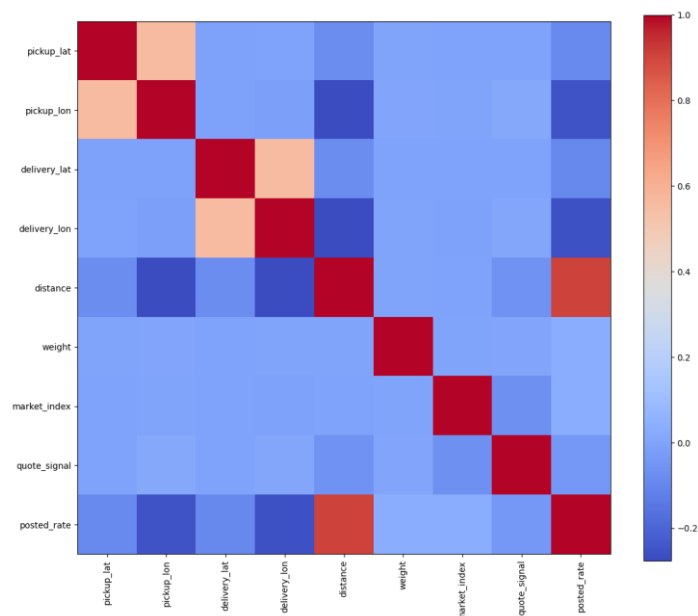


Figure 1: Correlation Heatmap

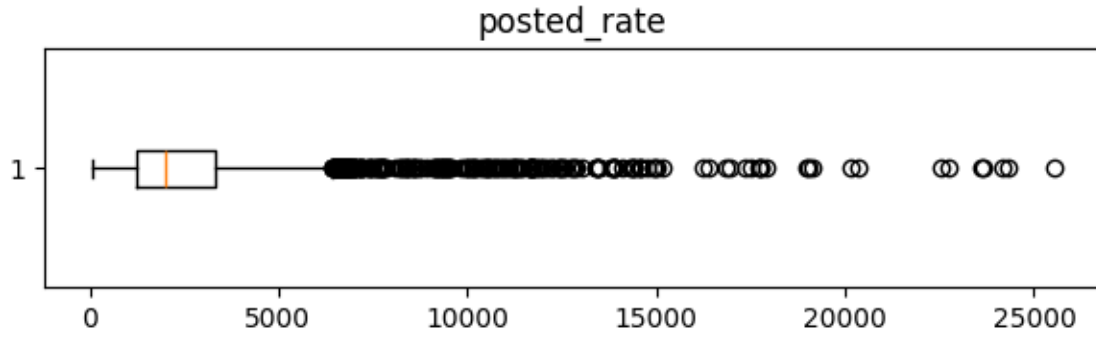


Figure 2: Boxplot of posted_rate

2. Train/ Testing Split (data_prepare.py)

The dataset (test-train.csv) was divided into:

- 80% Training
- 20% Testing

using a random train-validation split.

Table 1: Test/ Train Split

Dataset	Samples
Training	37,861
Testing	9,466

3. Training Strategy (model.py)

The neural network was trained using:

- Huber Loss
- Adam optimizer
- ReduceLROnPlateau learning-rate scheduler
- Early Stopping
- Best model checkpoint restoration

4. Model Network (model.py)

The architecture is given as:

- Three hidden layers (128, 64, 32) were used with ReLU, batch normalization, and dropout (0.2).

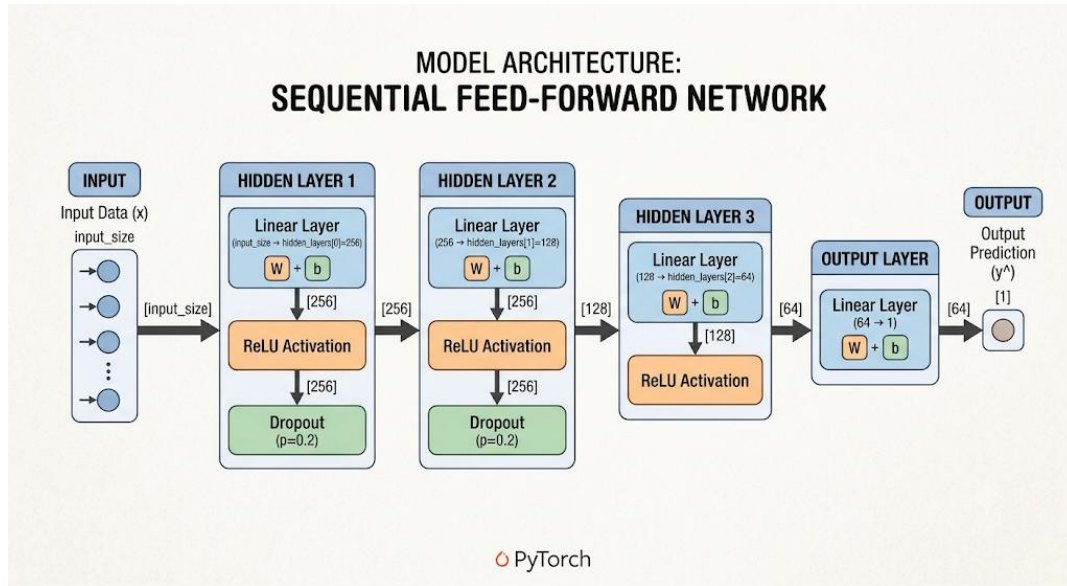


Figure 3: Model Architecture

5. Model Testing Performance Metrics

Model performance metrics are given as:

Table 2: Model Performance Metrics

Metric	Value
Loss	0.0104
RMSE	529.54
MAE	105.46
R^2	0.8689

6. December 2025 Prediction Chart

The supplied score.py script was used to validate both submission files and generate the required prediction chart.

The chart below shows the predicted freight rate for a fixed shipment:

- Pickup: Lexington
- Delivery: Fort Wayne
- Distance: 360 miles
- Equipment: Dry Van
- Weight: 32,000 lb

Only the shipment date changes from **December 1–31, 2025**.

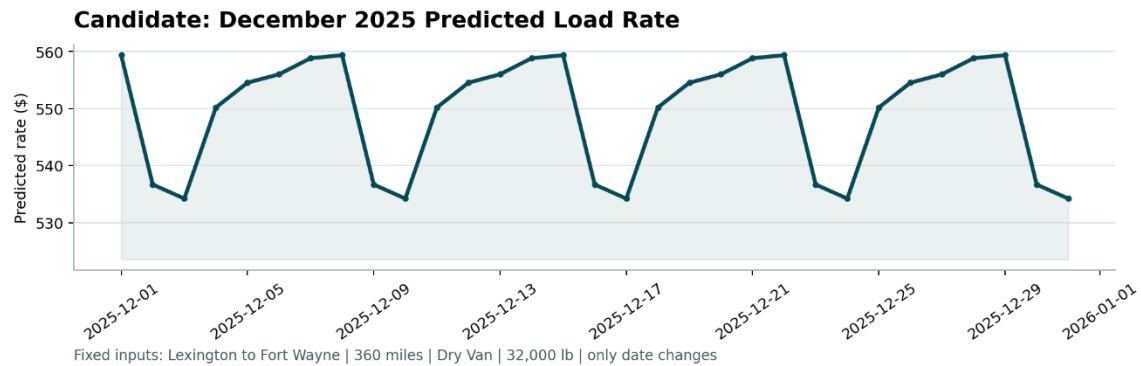


Figure 4: December 2025 Chart

7. Observations

- The model achieved an **R² score of 0.8689** on the validation set, indicating good predictive capability.
- Feature engineering using geographical coordinate differences (delta_lat, delta_lon) improved prediction accuracy.
- Applying a logarithmic transformation to the target variable stabilized training and reduced prediction error.
- Early stopping and adaptive learning rate scheduling improved model generalization.